

# VITYarthi - Build Your Own Project

## General Project Instructions & Submission Guidelines

As part of the flipped course evaluation, each student must build their **own original project**, aligned with the concepts covered in the subject syllabus.

Instead of a fixed project, you now have the freedom to propose, design, and build a project of your choice, provided it meets the academic and technical expectations described below.

Please note that the submitted project must be relevant to the course for which it is being submitted. Any irrelevant project will result in the submission being rejected and a grade of **0 marks** being awarded.

## 1. Project Objective

The goal of this project is to allow students to apply the subject concepts in a real-world context by:

- Identifying a meaningful problem
  - Designing a technical solution
  - Implementing the solution using the tools/methods learned in the course
  - Demonstrating understanding through documentation and evaluation
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## 2. Scope & Requirements

Your project must include:

## 2.1 Functional Requirements

These are the specific features or capabilities your project provides.

Your project must include at least:

- **Three major functional modules**
- Clear input/output structure
- A logical workflow of how the user interacts with the system

Examples (generic):

- User management
  - Data input & processing
  - Reporting or analytics
  - CRUD operations
  - Prediction or classification (for ML subjects)
  - Simulation or visualization (for relevant subjects)
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## 2.2 Non-Functional Requirements

You must specify at least **four (4)** non-functional requirements such as:

- Performance
- Security
- Usability

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- Reliability
  - Scalability
  - Maintainability
  - Error handling strategy
  - Logging or monitoring
  - Resource efficiency
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### 3. Technical Expectations

Depending on the subject, include the appropriate technical elements:

- Proper architectural design
- Correct application of subject concepts (algorithms, data structures, models, patterns, frameworks, tools, etc.)
- Modular and clean implementation
- Appropriate documentation and comments
- Validation and error handling
- Version control usage (Git)

If your subject involves coding or implementation:

- Minimum **5-10 meaningful modules/classes/files**
  - Proper folder or package structure
  - Testing wherever applicable (unit tests or validation tests)
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## 4. Design & Documentation Requirements

Your project must include the following design artefacts:

- **Problem Statement**
- **Objectives**
- **Functional Requirements**
- **Non-functional Requirements**
- **System Architecture Diagram**
- **Process Flow or Workflow Diagram**
- **UML Diagrams** (as applicable):
  - Use Case Diagram
  - Class Diagram / Component Diagram
  - Sequence Diagram
- **Database/Storage Design** (if applicable):
  - ER Diagram
  - Schema Design

If it is a computation-heavy or ML-related course, include:

- Dataset description
  - Model selection rationale
  - Evaluation methodology
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## 5. GitHub Repository Requirements

Each student must submit a **GitHub repository** containing:

### 5.1 **README.md**

This must clearly include:

- Project title
- Overview of the project
- Features
- Technologies/tools used
- Steps to install & run the project
- Instructions for testing
- Screenshots (optional but recommended)

### 5.2 **statement.md**

This file must contain:

- Problem statement
- Scope of the project
- Target users
- High-level features

### 5.3 **Source Code / Project Files**

Organized, modular, and complete.

Include data files, scripts, assets, or configuration files as needed.

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## 6. Project Report Submission (PDF on the Portal)

Along with the GitHub repository, you must upload a **detailed project report** in PDF format.

The report must include:

1. **Cover Page**
2. **Introduction**
3. **Problem Statement**
4. **Functional Requirements**
5. **Non-functional Requirements**
6. **System Architecture**
7. **Design Diagrams**
  - Use Case Diagram
  - Workflow Diagram
  - Sequence Diagram
  - Class/Component Diagram
  - ER Diagram (if storage used)
8. **Design Decisions & Rationale**
9. **Implementation Details**
10. **Screenshots / Results**
11. **Testing Approach**
12. **Challenges Faced**

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13. Learnings & Key Takeaways

14. Future Enhancements

15. References

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## 7. Evaluation Rubric

| Component                            | Weightage   |
|--------------------------------------|-------------|
| Problem Understanding & Requirements | 10%         |
| Design & Documentation               | 20%         |
| Implementation Quality               | 25%         |
| Innovation, Depth & Complexity       | 15%         |
| GitHub Repository & Version Control  | 10%         |
| Project Report                       | 20%         |
| <b>Total</b>                         | <b>100%</b> |